

SmartLink AI — Complete Frontend Guide

v2.0 — Register, Login, Dashboard, QR Code Display

Frontend Overview

Three HTML pages, all sharing the same CSS and dark navy theme:

File	Purpose	Auth required?
client/register.html	Create new account form	No
client/login.html	Login form, saves JWT to localStorage	No
client/index.html	Main dashboard — shorten, view links, analytics, QR	Yes — redirects to login if no token

Shared design decisions:

- All pages share identical CSS variables and color scheme — dark navy #0f0f1a background
- JWT token stored in localStorage under key: smartlink_token
- All protected API calls send: Authorization: Bearer header
- If no token found on index.html load — redirect to login.html immediately
- Fetch API used throughout — no external JS libraries needed

1. SHARED CSS (copy into all 3 pages)

```
<style>
* { box-sizing: border-box; margin: 0; padding: 0; }
body {
  font-family: 'Segoe UI', sans-serif;
  background: #0f0f1a;
  color: #e0e0e0;
  min-height: 100vh;
}
.navbar {
  background: #16213e;
  padding: 16px 32px;
  display: flex;
  align-items: center;
  justify-content: space-between;
  border-bottom: 1px solid #1a1a3e;
}
.navbar-brand { font-size: 20px; color: #4fc3f7; font-weight: 700; }
.navbar-right { display: flex; align-items: center; gap: 12px; }
.navbar-user { font-size: 13px; color: #aaa; }
.container { max-width: 720px; margin: 0 auto; padding: 24px 20px; }
.card {
  background: #16213e;
  border-radius: 12px;
  padding: 28px;
  border: 1px solid #1e2d50;
  margin-bottom: 20px;
}
label {
  display: block;
  font-size: 12px;
  color: #aaa;
  margin-bottom: 5px;
  margin-top: 14px;
}
```

```

label:first-child { margin-top: 0; }
input {
  width: 100%;
  padding: 11px 14px;
  border-radius: 8px;
  border: 1px solid #2a3a5e;
  background: #0f0f1a;
  color: #e0e0e0;
  font-size: 14px;
  outline: none;
  transition: border 0.2s;
}
input:focus { border-color: #4fc3f7; }
.row { display: flex; gap: 12px; }
.row > div { flex: 1; }
.btn {
  width: 100%;
  margin-top: 18px;
  padding: 13px;
  border: none;
  border-radius: 8px;
  background: linear-gradient(135deg, #4fc3f7, #0288d1);
  color: #fff;
  font-size: 15px;
  font-weight: 600;
  cursor: pointer;
  transition: opacity 0.2s;
}
.btn:hover { opacity: 0.88; }
.btn:disabled { opacity: 0.5; cursor: not-allowed; }
.btn-outline {
  width: 100%;
  padding: 11px;
  border: 1px solid #4fc3f7;
  border-radius: 8px;
  background: transparent;
  color: #4fc3f7;
  font-size: 14px;
  cursor: pointer;
  transition: background 0.2s;
  margin-top: 10px;
}
.btn-outline:hover { background: #1a2540; }
.btn-danger {
  padding: 6px 12px;
  border: 1px solid #ef5350;
  border-radius: 6px;
  background: transparent;
  color: #ef5350;
  font-size: 12px;
  cursor: pointer;
}
.btn-danger:hover { background: #ef535022; }
.error-msg { color: #ef5350; font-size: 13px; margin-top: 10px; display: none; }
.error-msg.show { display: block; }
.success-msg { color: #81c784; font-size: 13px; margin-top: 10px; display: none; }
.success-msg.show { display: block; }
.section-title {
  font-size: 12px;
  color: #aaa;
  margin-bottom: 10px;
  text-transform: uppercase;
  letter-spacing: 0.5px;
}
.hidden { display: none; }

```

```

    a { color: #4fc3f7; text-decoration: none; }
    a:hover { text-decoration: underline; }
</style>

```

2. client/register.html

```

<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8"/>
  <meta name="viewport" content="width=device-width, initial-scale=1.0"/>
  <title>SmartLink AI – Register</title>
  <!-- paste shared CSS here -->
</head>
<body>

<div class="navbar">
  <span class="navbar-brand">SmartLink AI</span>
</div>

<div class="container" style="max-width:420px; margin-top:60px;">
  <div class="card">
    <h2 style="font-size:20px;color:#fff;margin-bottom:4px;">Create Account</h2>
    <p style="font-size:13px;color:#aaa;margin-bottom:16px;">
      Already have an account? <a href="login.html">Login</a>
    </p>

    <label>Full Name</label>
    <input type="text" id="name" placeholder="Your name"/>

    <label>Email</label>
    <input type="email" id="email" placeholder="you@example.com"/>

    <label>Password (min 6 chars)</label>
    <input type="password" id="password" placeholder="....."/>

    <button class="btn" onclick="register()">Create Account</button>
    <p class="error-msg" id="errorMsg"></p>
    <p class="success-msg" id="successMsg"></p>
  </div>
</div>

<script>
  const BASE_URL = 'http://localhost:5000';

  // If already logged in, go to dashboard
  if (localStorage.getItem('smartlink_token')) {
    window.location.href = 'index.html';
  }

  async function register() {
    const name    = document.getElementById('name').value.trim();
    const email   = document.getElementById('email').value.trim();
    const password = document.getElementById('password').value;
    const btn     = document.querySelector('.btn');

    hideMessages();
    if (!name || !email || !password) {
      showError('All fields are required'); return;
    }

    btn.disabled = true;
    btn.textContent = 'Creating account...';

```

```

try {
  const res = await fetch(`${BASE_URL}/api/auth/register`, {
    method: 'POST',
    headers: { 'Content-Type': 'application/json' },
    body: JSON.stringify({ name, email, password })
  });
  const data = await res.json();

  if (!res.ok) { showError(data.error || 'Registration failed'); return; }

  // Save token and redirect
  localStorage.setItem('smartlink_token', data.token);
  localStorage.setItem('smartlink_user', JSON.stringify(data.user));
  showSuccess('Account created! Redirecting...');
  setTimeout(() => window.location.href = 'index.html', 1200);

} catch (err) {
  showError('Server error. Is the backend running?');
} finally {
  btn.disabled = false;
  btn.textContent = 'Create Account';
}
}

function showError(msg) {
  document.getElementById('errorMsg').textContent = msg;
  document.getElementById('errorMsg').classList.add('show');
}
function showSuccess(msg) {
  document.getElementById('successMsg').textContent = msg;
  document.getElementById('successMsg').classList.add('show');
}
function hideMessages() {
  document.getElementById('errorMsg').classList.remove('show');
  document.getElementById('successMsg').classList.remove('show');
}
}
</script>
</body>
</html>

```

3. client/login.html

```

<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8"/>
  <meta name="viewport" content="width=device-width, initial-scale=1.0"/>
  <title>SmartLink AI – Login</title>
  <!-- paste shared CSS here -->
</head>
<body>

<div class="navbar">
  <span class="navbar-brand">SmartLink AI</span>
</div>

<div class="container" style="max-width:420px; margin-top:60px;">
  <div class="card">
    <h2 style="font-size:20px;color:#fff;margin-bottom:4px;">Welcome Back</h2>
    <p style="font-size:13px;color:#aaa;margin-bottom:16px;">
      No account? <a href="register.html">Register</a>
    </p>

    <label>Email</label>

```

```

<input type="email" id="email" placeholder="you@example.com"/>

<label>Password</label>
<input type="password" id="password" placeholder="••••••••"
      onkeydown="if(event.key==='Enter') login()"/>

<button class="btn" onclick="login()">Login</button>
<p class="error-msg" id="errorMsg"></p>
</div>
</div>

<script>
const BASE_URL = 'http://localhost:5000';

if (localStorage.getItem('smartlink_token')) {
  window.location.href = 'index.html';
}

async function login() {
  const email = document.getElementById('email').value.trim();
  const password = document.getElementById('password').value;
  const btn = document.querySelector('.btn');
  const errorMsg = document.getElementById('errorMsg');

  errorMsg.classList.remove('show');
  if (!email || !password) { showError('Both fields required'); return; }

  btn.disabled = true;
  btn.textContent = 'Logging in...';

  try {
    const res = await fetch(`${BASE_URL}/api/auth/login`, {
      method: 'POST',
      headers: { 'Content-Type': 'application/json' },
      body: JSON.stringify({ email, password })
    });
    const data = await res.json();

    if (!res.ok) { showError(data.error || 'Login failed'); return; }

    localStorage.setItem('smartlink_token', data.token);
    localStorage.setItem('smartlink_user', JSON.stringify(data.user));
    window.location.href = 'index.html';

  } catch (err) {
    showError('Server error. Is the backend running?');
  } finally {
    btn.disabled = false;
    btn.textContent = 'Login';
  }
}

function showError(msg) {
  document.getElementById('errorMsg').textContent = msg;
  document.getElementById('errorMsg').classList.add('show');
}
</script>
</body>
</html>

```

4. client/index.html — Main Dashboard

This is the main app page. It checks for token on load, shows the shorten form, my links list, QR codes, and analytics.

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8"/>
  <meta name="viewport" content="width=device-width, initial-scale=1.0"/>
  <title>SmartLink AI</title>
  <!-- paste shared CSS here -->
  <style>
    /* Extra styles for dashboard */
    .short-url-box {
      display: flex; align-items: center;
      background: #0f0f1a; border: 1px solid #2a3a5e;
      border-radius: 8px; padding: 12px 14px;
      gap: 10px; margin-bottom: 14px;
    }
    .short-url-box a { flex:1; color:#4fc3f7; font-size:15px; word-break:break-all; }
    .copy-btn {
      padding: 6px 14px; background: #1e2d50;
      color: #aaa; border: none; border-radius: 6px;
      cursor: pointer; font-size: 12px;
    }
    .copy-btn:hover { background:#2a3a5e; color:#fff; }
    .ai-meta {
      background: #1a2540; border-left: 3px solid #4fc3f7;
      border-radius: 6px; padding: 12px 16px; margin-bottom: 14px;
    }
    .ai-meta h4 { font-size:14px; color:#fff; margin-bottom:4px; }
    .ai-meta p { font-size:12px; color:#aaa; }
    .qr-box { text-align:center; margin-bottom:14px; }
    .qr-box img { width:160px; height:160px; border-radius:8px; border:2px solid #1e2d50; }
    .qr-box a {
      display:inline-block; margin-top:8px; font-size:12px;
      color:#4fc3f7; cursor:pointer;
    }
    .stat-cards { display:flex; gap:12px; margin-bottom:16px; }
    .stat {
      flex:1; background:#1a2540; border-radius:10px;
      padding:16px; text-align:center; border:1px solid #1e2d50;
    }
    .stat .num { font-size:24px; font-weight:700; color:#4fc3f7; }
    .stat .lbl { font-size:11px; color:#888; margin-top:4px; }
    .date-table { width:100%; border-collapse:collapse; font-size:13px; margin-bottom:16px; }
    .date-table th { text-align:left; padding:8px 10px; background:#1a2540; color:#aaa; }
    .date-table td { padding:8px 10px; border-bottom:1px solid #1e2d50; color:#ddd; }
    .ai-summary-box {
      background:#1a2540; border-left:3px solid #81c784;
      border-radius:6px; padding:14px 16px;
    }
    .ai-summary-box h4 { font-size:13px; color:#81c784; margin-bottom:6px; }
    .ai-summary-box p { font-size:13px; color:#ccc; line-height:1.6; }
    .link-item {
      background:#0f0f1a; border:1px solid #1e2d50;
      border-radius:8px; padding:12px 14px; margin-bottom:10px;
      display:flex; align-items:center; justify-content:space-between; gap:10px;
    }
    .link-item-info { flex:1; min-width:0; }
    .link-item-title { font-size:13px; color:#fff; margin-bottom:2px; white-space:nowrap; overflow:hidden; text-overfl
    .link-item-url { font-size:11px; color:#4fc3f7; }
    .link-item-meta { font-size:11px; color:#666; margin-top:2px; }
    .link-actions { display:flex; gap:8px; flex-shrink:0; }
  </style>
</head>
<body>

<div class="navbar">
```

```

<span class="navbar-brand">SmartLink AI</span>
<div class="navbar-right">
  <span class="navbar-user" id="navUser"></span>
  <button onclick="logout()" style="padding:6px 14px;background:#1e2d50;color:#aaa;
    border:none;border-radius:6px;cursor:pointer;font-size:12px;">Logout</button>
</div>
</div>

<div class="container">

  <!-- SHORTEN FORM -->
  <div class="card">
    <h3 style="color:#fff;margin-bottom:16px;font-size:16px;">Shorten a URL</h3>
    <label>Paste your long URL *</label>
    <input type="url" id="originalUrl" placeholder="https://example.com/very/long/url"/>
    <div class="row">
      <div>
        <label>Custom Alias (optional)</label>
        <input type="text" id="customAlias" placeholder="e.g. mylink"/>
      </div>
      <div>
        <label>Expires in days (optional)</label>
        <input type="number" id="expiresIn" placeholder="e.g. 7" min="1"/>
      </div>
    </div>
    <button class="btn" id="shortenBtn" onclick="shortenUrl()">Shorten URL</button>
    <p class="error-msg" id="errorMsg"></p>
  </div>

  <!-- RESULT CARD -->
  <div class="card hidden" id="resultCard">
    <p class="section-title">Your Short Link</p>
    <div class="short-url-box">
      <a href="#" id="shortUrlLink" target="_blank"></a>
      <button class="copy-btn" onclick="copyUrl()">Copy</button>
    </div>
    <div class="ai-meta">
      <h4 id="aiTitle"></h4>
      <p id="aiDesc"></p>
    </div>
    <div class="qr-box">
      <img id="qrImage" src="" alt="QR Code"/>
      <br/>
      <a onclick="downloadQR()">Download QR Code</a>
    </div>
    <button class="btn-outline" onclick="loadAnalytics()">View Analytics</button>
  </div>

  <!-- ANALYTICS CARD -->
  <div class="card hidden" id="analyticsCard">
    <p class="section-title">Link Analytics</p>
    <div class="stat-cards">
      <div class="stat"><div class="num" id="totalClicks">0</div><div class="lbl">Total Clicks</div></div>
      <div class="stat"><div class="num" id="topDevice"></div><div class="lbl">Top Device</div></div>
      <div class="stat"><div class="num" id="topBrowser"></div><div class="lbl">Top Browser</div></div>
    </div>
    <p class="section-title">Clicks by Date</p>
    <table class="date-table">
      <thead><tr><th>Date</th><th>Clicks</th></tr></thead>
      <tbody id="dateTableBody"></tbody>
    </table>
    <div class="ai-summary-box">
      <h4>AI Summary</h4>
      <p id="aiSummaryText">Loading...</p>
    </div>
  </div>

```

```

</div>

<!-- MY LINKS -->
<div class="card">
  <div style="display:flex;justify-content:space-between;align-items:center;margin-bottom:14px;">
    <p class="section-title" style="margin:0;">My Links</p>
    <button onclick="loadMyLinks()" style="padding:5px 12px;background:#1e2d50;
      color:#aaa;border:none;border-radius:6px;cursor:pointer;font-size:12px;">Refresh</button>
  </div>
  <div id="myLinksList"><p style="color:#666;font-size:13px;">Loading your links...</p></div>
</div>

</div><!-- end container -->

<script>
  const BASE_URL = 'http://localhost:5000';
  let token = localStorage.getItem('smartlink_token');
  let currentShortCode = '';
  let currentQRData = '';

  // Auth guard
  if (!token) window.location.href = 'login.html';

  // Show username in navbar
  const user = JSON.parse(localStorage.getItem('smartlink_user') || '{}');
  if (user.name) document.getElementById('navUser').textContent = user.name;

  // Auth header helper
  const authHeaders = () => ({
    'Content-Type': 'application/json',
    'Authorization': `Bearer ${token}`
  });

  // Load links on page load
  window.addEventListener('load', loadMyLinks);

  // ■■ SHORTEN ■■
  async function shortenUrl() {
    const originalUrl = document.getElementById('originalUrl').value.trim();
    const customAlias = document.getElementById('customAlias').value.trim();
    const expiresIn = document.getElementById('expiresIn').value.trim();
    const btn = document.getElementById('shortenBtn');
    const errorMsg = document.getElementById('errorMsg');

    errorMsg.classList.remove('show');
    if (!originalUrl) { showError('Please enter a URL'); return; }

    btn.disabled = true;
    btn.textContent = 'Shortening...';

    try {
      const body = { originalUrl };
      if (customAlias) body.customAlias = customAlias;
      if (expiresIn) body.expiresIn = parseInt(expiresIn);

      const res = await fetch(`${BASE_URL}/api/shorten`, {
        method: 'POST', headers: authHeaders(), body: JSON.stringify(body)
      });
      const data = await res.json();

      if (res.status === 401) { logout(); return; }
      if (!res.ok) { showError(data.error || 'Something went wrong'); return; }

      currentShortCode = data.shortCode;
      currentQRData = data.qrCode;
    }
  }

```



```

document.getElementById('shortUrlLink').href      = data.shortUrl;
document.getElementById('shortUrlLink').textContent = data.shortUrl;
document.getElementById('aiTitle').textContent   = data.title;
document.getElementById('aiDesc').textContent    = data.description;
document.getElementById('qrImage').src          = data.qrCode;

document.getElementById('resultCard').classList.remove('hidden');
document.getElementById('analyticsCard').classList.add('hidden');
loadMyLinks();

} catch (err) {
  showError('Server error. Is the backend running?');
} finally {
  btn.disabled = false;
  btn.textContent = 'Shorten URL';
}
}

// ■■ ANALYTICS ■■
async function loadAnalytics() {
  if (!currentShortCode) return;
  document.getElementById('analyticsCard').classList.remove('hidden');
  document.getElementById('aiSummaryText').textContent = 'Loading AI summary...';

  try {
    const res = await fetch(`${BASE_URL}/api/analytics/${currentShortCode}`,
      { headers: authHeaders() });
    const data = await res.json();

    document.getElementById('totalClicks').textContent = data.totalClicks || 0;
    const devices = data.devices || {};
    const browsers = data.browsers || {};
    document.getElementById('topDevice').textContent =
      Object.keys(devices).sort((a,b) => devices[b] - devices[a])[0] || '-';
    document.getElementById('topBrowser').textContent =
      Object.keys(browsers).sort((a,b) => browsers[b] - browsers[a])[0] || '-';

    const tbody = document.getElementById('dataTableBody');
    const byDate = data.byDate || {};
    tbody.innerHTML = '';
    const sorted = Object.entries(byDate).sort((a,b) => b[0].localeCompare(a[0]));
    if (sorted.length === 0) {
      tbody.innerHTML = '<tr><td colspan="2" style="color:#666">No clicks yet</td></tr>';
    } else {
      sorted.forEach(([date, count]) => {
        tbody.innerHTML += `<tr><td>${date}</td><td>${count}</td></tr>`;
      });
    }
    document.getElementById('aiSummaryText').textContent =
      data.aiSummary || 'No summary available yet.';

  } catch (err) {
    document.getElementById('aiSummaryText').textContent = 'Failed to load analytics.';
  }
}

// ■■ MY LINKS ■■
async function loadMyLinks() {
  const list = document.getElementById('myLinksList');
  try {
    const res = await fetch(`${BASE_URL}/api/my-links`, { headers: authHeaders() });
    const data = await res.json();

    if (res.status === 401) { logout(); return; }
  }
}

```

```

    if (!data.urls || data.urls.length === 0) {
      list.innerHTML = '<p style="color:#666;font-size:13px;">No links yet. Shorten one above!</p>';
      return;
    }

    list.innerHTML = data.urls.map(url => `
      <div class="link-item">
        <div class="link-item-info">
          <div class="link-item-title">${url.title || 'Untitled'}</div>
          <div class="link-item-url">
            <a href="${BASE_URL}/${url.shortCode}" target="_blank">
              ${BASE_URL}/${url.shortCode}
            </a>
          </div>
          <div class="link-item-meta">
            ${url.clicks ? url.clicks.length : 0} clicks &nbsp;  ·&nbsp;  
            ${new Date(url.createdAt).toLocaleDateString()}
            ${url.expiresAt ? ' &nbsp;  ·&nbsp;   Expires ' + new Date(url.expiresAt).toLocaleDateString() : ''}
          </div>
        </div>
        <div class="link-actions">
          <button class="copy-btn" onclick="navigator.clipboard.writeText('${BASE_URL}/${url.shortCode}')">Copy</button>
          <button class="btn-danger" onclick="deleteLink('${url.shortCode}')">Delete</button>
        </div>
      </div>
    `).join('');

  } catch (err) {
    list.innerHTML = '<p style="color:#ef5350;font-size:13px;">Failed to load links.</p>';
  }
}

// ■■ DELETE LINK ■■
async function deleteLink(shortCode) {
  if (!confirm('Delete this link?')) return;
  try {
    const res = await fetch(`${BASE_URL}/api/url/${shortCode}`, {
      method: 'DELETE', headers: authHeaders()
    });
    if (res.ok) loadMyLinks();
  } catch (err) { alert('Delete failed'); }
}

// ■■ QR DOWNLOAD ■■
function downloadQR() {
  if (!currentQRData) return;
  const a = document.createElement('a');
  a.href = currentQRData;
  a.download = `qr-${currentShortCode}.png`;
  a.click();
}

// ■■ COPY URL ■■
function copyUrl() {
  const url = document.getElementById('shortUrlLink').textContent;
  navigator.clipboard.writeText(url).then(() => {
    const btn = document.querySelector('.copy-btn');
    btn.textContent = 'Copied!';
    setTimeout(() => btn.textContent = 'Copy', 2000);
  });
}

function showError(msg) {
  const el = document.getElementById('errorMsg');
  el.textContent = msg;
}
```

```

    el.classList.add('show');
  }

  function logout() {
    localStorage.removeItem('smartlink_token');
    localStorage.removeItem('smartlink_user');
    window.location.href = 'login.html';
  }
</script>
</body>
</html>

```

5. .env.example

```

MONGO_URI=mongodb://localhost:27017/smartlink
REDIS_URL=redis://localhost:6379
GROQ_API_KEY=your_groq_key_here
SAFE_BROWSING_API_KEY=your_google_safe_browsing_key_here
JWT_SECRET=replace_with_long_random_string_minimum_32_characters
JWT_EXPIRES_IN=7d
BASE_URL=http://localhost:5000
PORT=5000

```

6. README.md

SmartLink AI v2.0

AI-powered URL shortener with user authentication, QR code generation, and real-time click analytics.

Features

- User registration and login with JWT authentication
- Shorten any URL with optional custom alias and expiry
- QR code generated automatically for every short link
- AI-generated title and description for every link (Groq Llama 3.1)
- Click analytics: device, browser, and date tracking
- AI-generated plain English analytics summary
- Redis caching for fast redirects (~1ms response time)
- Rate limiting (10 requests/IP/minute)
- Malicious URL detection via Google Safe Browsing API
- Each user can only see and manage their own links

Tech Stack

Backend: Node.js, Express.js
 Database: MongoDB (Mongoose)
 Cache: Redis (ioredis)
 Auth: bcryptjs + jsonwebtoken
 AI: Groq API (Llama 3.1-8b-instant)
 QR Code: qrcode npm package

Getting Started

Prerequisites

- Node.js v18+
- MongoDB (local or Atlas)
- Redis (local or Upstash)
- Groq API key → console.groq.com (free)
- Safe Browsing key → console.developers.google.com (free)

Installation

```

git clone https://github.com/yourusername/smartlink-ai
cd smartlink-ai
npm install

```

```
cp .env.example .env
# Fill in all values in .env
npm run dev

### Open the app
Open client/register.html in browser to create an account
Then you will be redirected to the main dashboard

## API Endpoints
POST   /api/auth/register      Register new user
POST   /api/auth/login      Login, receive JWT token
GET    /api/auth/me          Get logged in user (protected)
POST   /api/shorten        Create short URL (protected)
GET    /api/my-links        Get all my links (protected)
DELETE /api/url/:shortCode   Delete my link (protected)
GET    /api/analytics/:shortCode Get analytics + AI summary (protected)
GET    /:shortCode          Public redirect to original URL

## Deployment
Backend:  Railway or Render
Database: MongoDB Atlas (free tier)
Redis:    Upstash (free tier, works perfectly with ioredis)
Frontend: Vercel or Netlify (drag and drop client/ folder)

## Author
[Your Name] | [GitHub] | [LinkedIn]
```

7. HOW TO RUN LOCALLY — STEP BY STEP

1. Install Node.js v18+ from nodejs.org
2. Install MongoDB Community Edition — start it with: `mongod`
3. Install Redis — start it with: `redis-server`
4. Get free Groq API key at console.groq.com — takes 2 minutes
5. Get free Google Safe Browsing key at console.developers.google.com
6. Clone/create project folder with the structure from backend guide
7. Run: `npm install` — installs all packages
8. Copy `.env.example` to `.env` — fill in all 8 values
9. Run: `npm run dev` — nodemon starts server on port 5000
10. Open `client/register.html` in VS Code with Live Server extension
11. Register an account → auto-redirected to dashboard
12. Paste any URL, click Shorten — see short link + QR code appear
13. Click View Analytics to see AI summary

8. COMMON ERRORS AND FIXES

Error	Cause	Fix
Cannot use import	nanoid v4	<code>npm install nanoid@3</code>
401 on all requests	Token missing/expired	Login again, check <code>localStorage</code>
CORS error	<code>cors</code> not installed	<code>npm install cors</code> , add <code>app.use(cors())</code>
Redis ECONNREFUSED	Redis not running	Run <code>redis-server</code> or use Upstash
QR not showing	<code>base64</code> not prefixed	Check <code>qrCode</code> starts with <code>data:image/png</code>
JWT invalid	Wrong <code>JWT_SECRET</code>	Check <code>.env</code> <code>JWT_SECRET</code> matches exactly
